

Tanner O'Rourke / ML Research Engineer

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Left a senior engineering role to complete an MS in AI full-time. 5+ years of production engineering, with a background in HCI and psychology. Specializes in model evaluation and interpretability.

SKILLS

ML, Evaluation & Interpretability: PyTorch, Experiment Design, Behavioral Evaluation, LLM Evaluation, LLM-as-a-Judge, Confound Diagnostics, Bootstrap CIs, Post-Training (SFT, RFT, DPO), Model Probing, SAEs, Activation Patching, TransformerLens, Manifold Analysis, Tokenization

Infrastructure & Data: Agentic Workflows, MCP, ETL Pipelines, BigQuery, GraphQL, Docker, AWS S3, Memory-Efficient Training, Hugging Face Transformers/Datasets, Pandas, NumPy, xarray

Languages: Python, JavaScript, React, TypeScript, Next.js, C++, SQL

SELECT RESEARCH

Probing Temporal Compositionality in Self-Supervised Clinical Encoders [\[Code\]](#)

- Trained a five-size 11M to 216M encoder ladder under stop-gradient, EMA, and supervised objectives on ~327k MIMIC-IV encounters (~55k patients), with VICReg regularization, collapse monitoring, and live representation health monitoring.
- Testing whether temporal patient states compose from clinical features via linear probes, TopK SAEs, activation patching, and manifold geometry (PCA, PHATE, CKA); a proof-based agentic workflow rejects claims lacking seed and dimension agreement.

Cross-Format Representational Similarity Measures Output Form, Not Knowledge [\[Code\]](#) [\[Paper\]](#)

- Built a TransformerLens pipeline capturing residual-stream shifts on Gemma-2-2B across MCQ and open-ended renderings of 500 stratified MMLU questions, four prompt formats, under 0-shot and 5-shot elicitation.
- Falsified the metric's own construct validity: ~98% insensitive to answer-option content), template-dominated, with a primary AUC swinging 0.70 to 0.53 on the scoring rule alone; at chance on MCQ-correct questions where format exploitation is defined.

Predicting Wildfire Ignition with Multi-Modal Spatiotemporal ConvFormers [\[Code\]](#) [\[Datasets\]](#)

- Engineered a geospatial pipeline harmonizing ten raster and vector products (PRISM, AORC, MODIS, LANDFIRE, NLCD, lightning, WUI, roads, fire records) into a 38-channel daily datacube of Washington State (2003-2020), published to HuggingFace.
- Designed a two-path spatiotemporal ConvFormer: per-group CNN stems, spatial (windowed), channel, and temporal attention, FiLM terrain conditioning, trained as a proper scoring rule with negative subsampling and exact logit-shift correction.

NanoCoder: Building a Code Language Model From Scratch [\[Playlist\]](#)

- Built a code LM from-scratch (no AutoTokenizer, transformers, or trl): custom BPE tokenizer, 123M LLaMA-style decoder (RoPE/RMSNORM/SwiGLU/QK-norm), and execution-verified SFT/RFT/DPO, published as a five-part Colab walkthrough.

ENGINEERING EXPERIENCE

Senior Software Engineer | DirecTV (via Cognizant)

October 2023 - August 2025 | Promoted from Software Engineer in 3 months | Q3 '24 top IC across customer-facing web teams

- Technical lead for directv.com, drove implementation strategy with PM's, UX, and VP-level stakeholders. Mentored junior engineers and delivered high-profile releases to tens of millions of users.
- Designed CI/CD automations and modernized the core brand component library, enabling dependency consistency across department teams.
- Diagnosed rendering bottlenecks on DirecTV's 2nd highest-traffic page: 149% faster load time, 37% smaller bundle, 93% reduction in interaction latency.
- Independently introduced a lower environment component preview sandbox, adopted *org wide*.

Interactive Developer | Warner Bros. (via Cognizant)

October 2020 - October 2023

- Directed an 8-person development team building seven casting and scripting apps, while serving as sole day-to-day technical voice for the client team, owning UI decisions, implementation, and PR review.

EDUCATION

Master of Science: Artificial Intelligence | UT Austin | 2025-2026 (Completed in 3 semesters)

Focus: Evaluation, NLP, Applied ML Systems

Bachelor of Science: Computer Science | CU Boulder | 2016-2020 (Chancellor's Scholar)

Focus: Human-Centered Computing